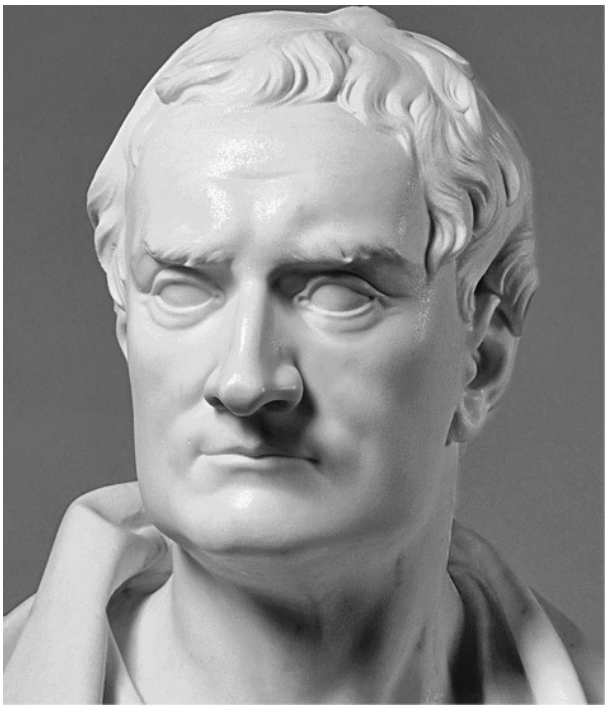
enabled the construction of modern physics—a physics that works incredibly well. And it assumes that time exists as an entity that runs in a way which is uniform and imperturbable. Newton writes equations containing the letter *t* for time that describe how things move *in time*.[47] What does this letter mean? Does *t* indicate time shaped by the longer hours of summer and the shorter ones of winter? Obviously not. It indicates time that is "absolute, true, and mathematical," assumed by Newton to run *independently of things that change or things that move*.



Newton: There is a time that passes even when nothing changes.

Clocks, for Newton, are devices that seek, albeit in a manner that is always imprecise, to follow this equal and uniform flowing of time. Newton writes that this "absolute, true, and mathematical" time is not perceptible. It must be deduced, through calculation and observation, from the regularity of phenomena. Newton's time is not the evidence given to us by our senses: it is an elegant intellectual construction. If, my dear cultivated reader, the existence of this Newtonian concept of time which is independent of things seems to you simple and natural, it's because you encountered it at school. Because it has gradually become the way in which we all think about time. It has filtered through school textbooks throughout the world and ended up becoming our common way of understanding time. We have turned it into our common sense. But the existence of a time that is uniform, independent of